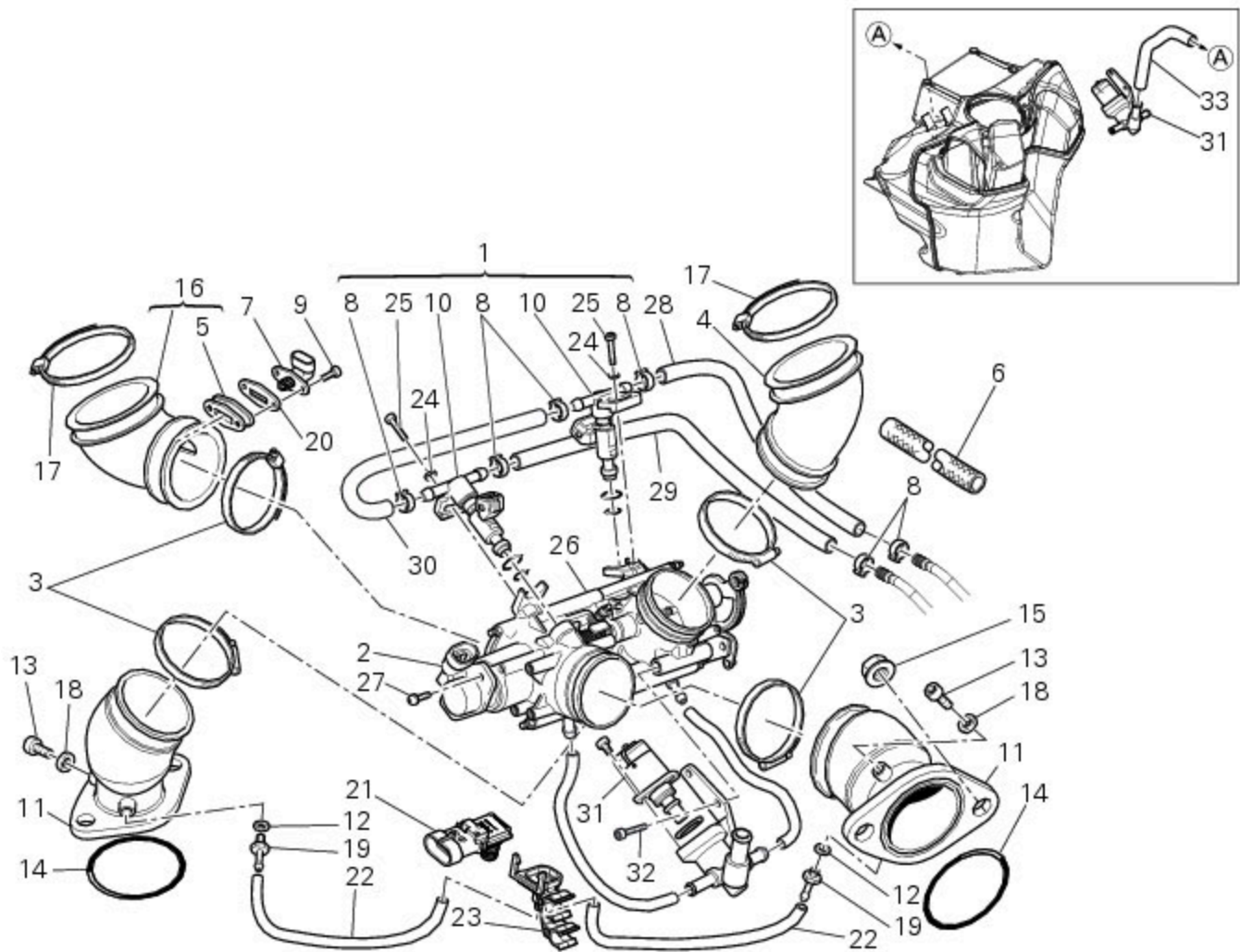


6 - Throttle body



- 1 Fuel system assembly
- 2 Throttle position sensor
- 3 Clamp
- 4 Hose
- 5 Rubber pad
- 6 Heat shrink sleeve
- 7 Air temperature sensor
- 8 Clamp
- 9 Screw
- 10 Injector
- 11 Intake manifold
- 12 Aluminium gasket
- 13 Screw
- 14 O-ring
- 15 Nut
- 16 Hose
- 17 Clamp
- 18 Washer
- 19 Union
- 20 Sealing washer
- 21 Pressure sensor
- 22 Hose
- 23 Sensor support
- 24 Washer
- 25 Screw
- 26 Throttle body assembly

- 27 Screw
- 28 Fuel feed hose
- 29 Fuel return hose
- 30 Connection hose
- 31 Stepper motor
- 32 Screw
- 33 Hose

i Spare parts catalogue

- 1100 ABS [INTAKE MANIFOLDS](#)
- 1100 ABS [AIR INTAKE - OIL BREATHER](#)
- 1100 S ABS [INTAKE MANIFOLDS](#)
- 1100 S ABS [AIR INTAKE - OIL BREATHER](#)

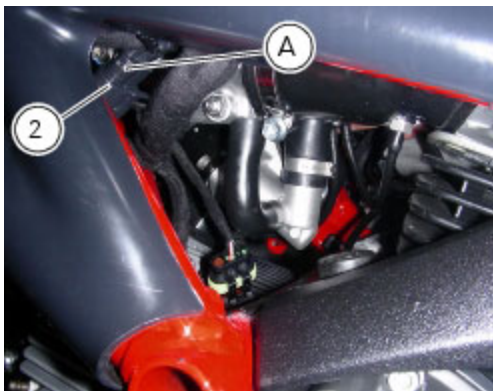
● Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but can be found in the exploded view diagram.

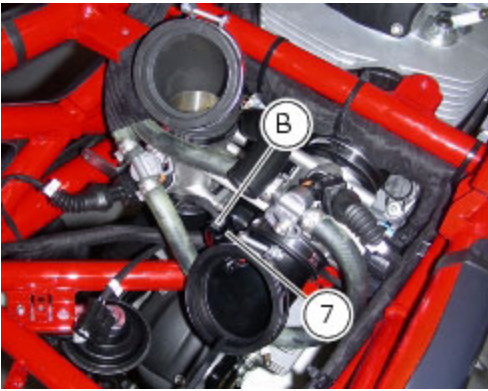
Removal of the throttle body

Operation	Section reference
Remove the seat	E 3, Removal of the seat
Remove the tank covers	E 2, Removal of the fuel tank fairings
Remove the fuel tank	L 2, Removal of the fuel tank
Remove the ABS control unit	G 9.2, Removing of the ABS control unit
Remove the battery	P 2, Battery support
Remove the ECU connector	M 3, Removal of the electronic control unit
Connect the starter motor/solenoid cable	P 3, Removal of the starter motor
Remove the airbox	L 7, Removal of the airbox
Disconnect the throttle cable	F 1, Removal of the throttle twistgrip

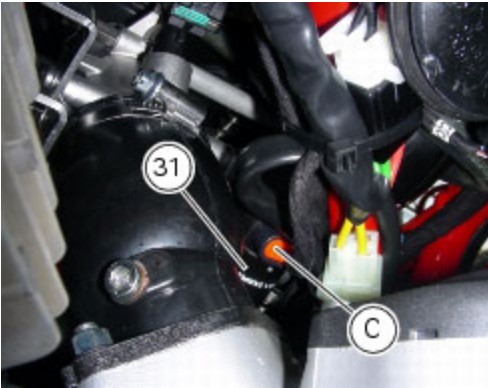
Disconnect the connector (A) connecting the main wiring harness to the throttle position sensor (2).



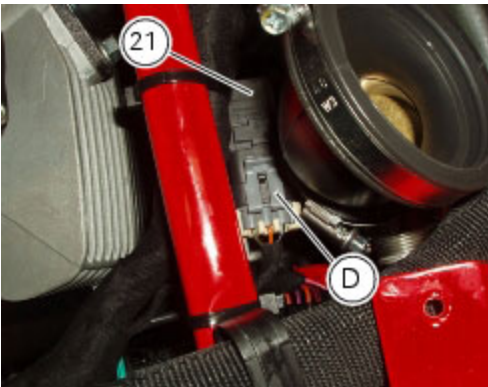
Disconnect the wiring connector (B) connecting the main wiring harness to the air temperature sensor (7).



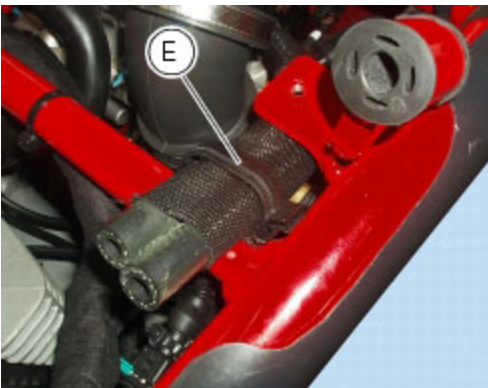
Disconnect the connector (C) connecting the main wiring loom to the stepper motor (31).



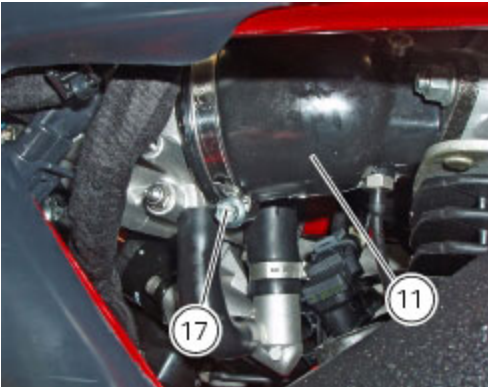
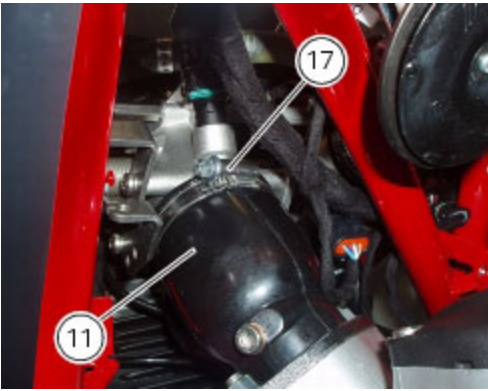
Disconnect the air pressure sensor (21) from the throttle body and detach connector (D) from the main wiring harness.



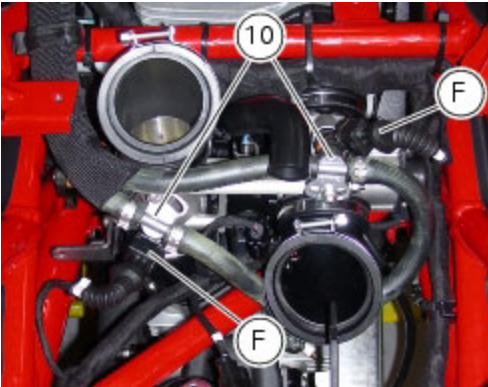
Release the clip clamp (E) and remove the delivery and return hoses from the frame.



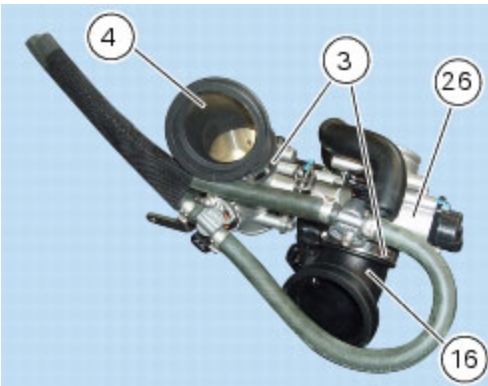
Loosen the clips (17) securing the throttle body to the intake manifolds (11).



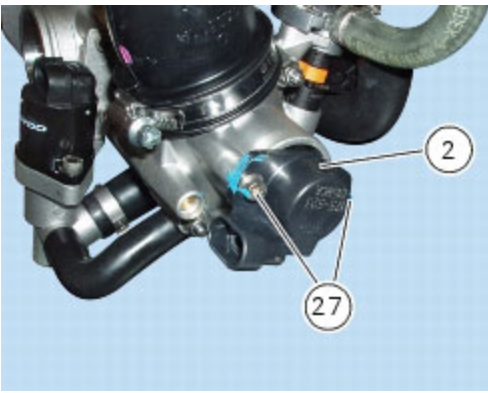
Disconnect the wiring connectors (F) connecting the main wiring loom to the injectors (10).



Remove the throttle body assembly (26) from the motorcycle.
If necessary, slacken off the clamps (3) securing the throttle body to the hoses (4) and (16) and remove the hoses.



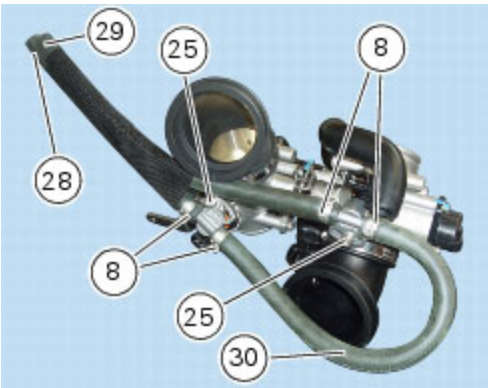
Undo the two screws (27) to remove the throttle position sensor (2) from the throttle body.



Removal of the fuel injectors

Undo the injector retaining clips (8) with the fuel feed hose (28), fuel return hose (29) and injector connection hose (30).

Undo the screws (25) to remove the injectors from the throttle body, recovering the washers (24).

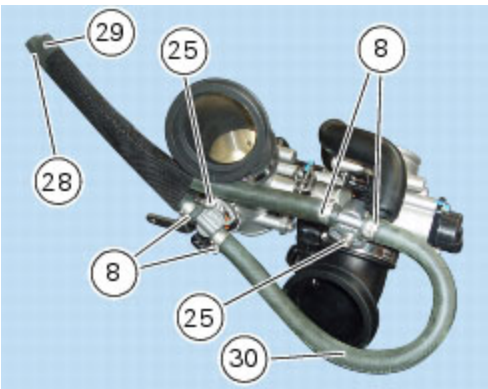


Refitting the injectors

Lubricate the injector O-rings.

Refit on injectors: fuel feed hose (28), fuel return hose (29) and injector connection hose (30), then tighten them using ties (8).

Fit washers (24) and tighten screws (25) to the specified torque (Sect. C 3, [Engine torque settings](#)).



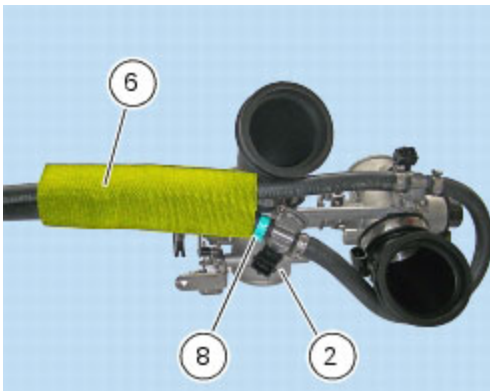
If the heat shrink sleeve (6) is to be renewed, slide it over hoses (28) and (29) and seat tight against the clamp (8), as shown in the photo.

Set the hot air blower to **150 °C**. While holding the pipes curved, apply heat to "shrink" the sleeve, holding the blower approximately **15 mm** from the sleeve. Pass the hot air blower along the sleeve several times (moving continuously from one end to the other) avoiding heating the same point for a prolonged time.



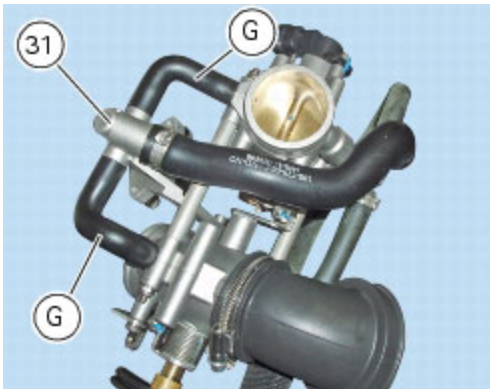
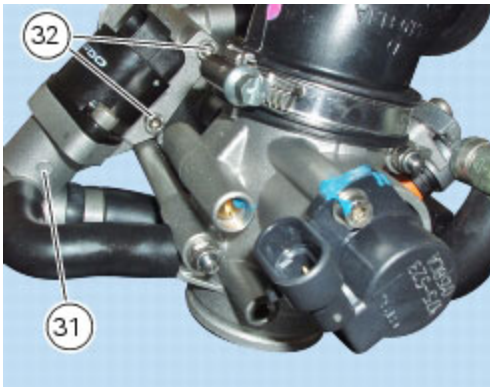
Important

The sleeve should only be heated for the timer it takes to "shrink"; do not carry on heating it beyond this point.



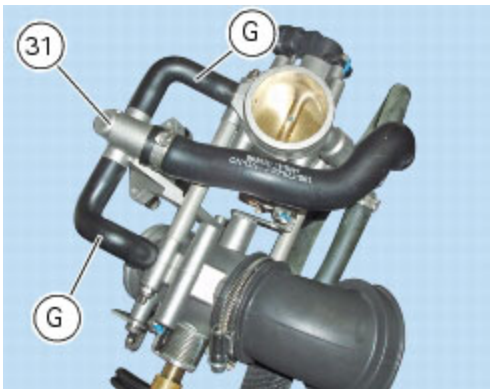
Removal of the stepper motor

The stepper motor (31) is installed in the central part of the throttle body:
 To remove it, unscrew the two screws (32) and disconnect the hoses (G) from the throttle body.
 Refer to Section M 3, [Stepper motor](#) for details on its operation.

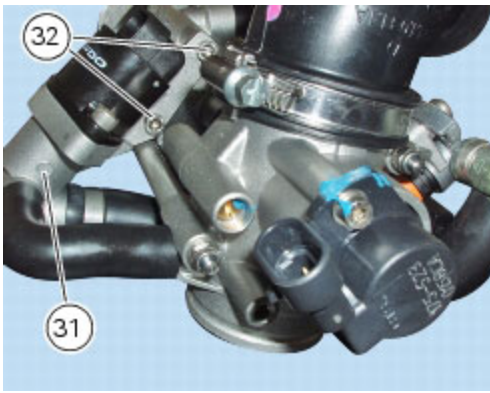


Refitting the stepper motor

Connect the hoses (G) to the throttle body.



Locate the stepper motor (31) with its support on the throttle body, fitting the screws (32).
 Tighten the screws (32).



Refitting the throttle body

If the hoses (4) and (16) were removed, reconnect them to the throttle body with the clamps (3), positioning the hoses and clamps as shown in the figure.



Note

The air temperature sensor (7) is installed on the vertical cylinder manifold (16); refer to Sect. M 3, [Air pressure sensor](#) for its installation.



Warning

The hose (16) which mounts the air temperature sensor (17) should be installed on the same side of the throttle body as the position sensor (2).

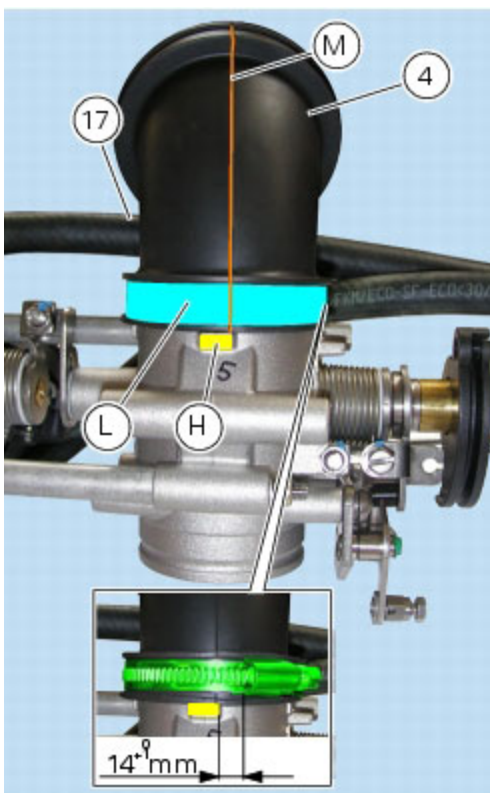
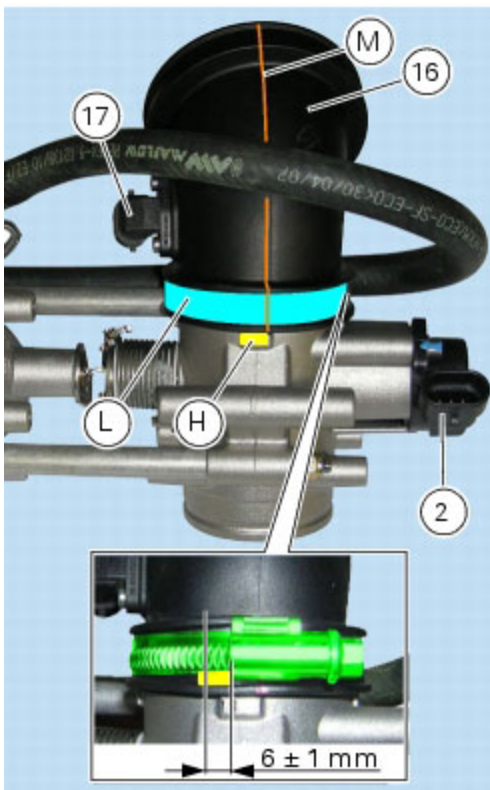


Note

The end of the hose with the smaller diameter is to be connected to the throttle body.

Take couplings (4) and (16) fully home on throttle body, ensuring the mould parting lines (M) of couplings match with the right-hand edge of throttle body sockets (H), as shown below.

Tighten the clamps to the specified torque (Sect. C 3, [Frame torque settings](#)), complying with the positioning shown in the figure and ensuring they do not pop out of their seats (L) on couplings.

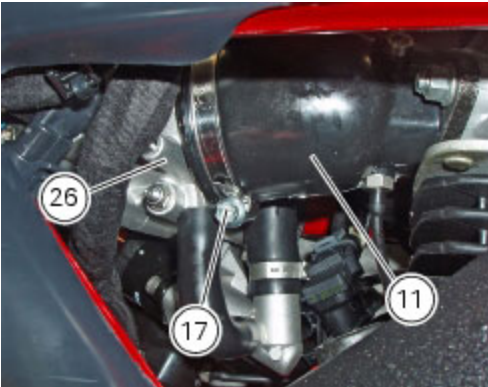
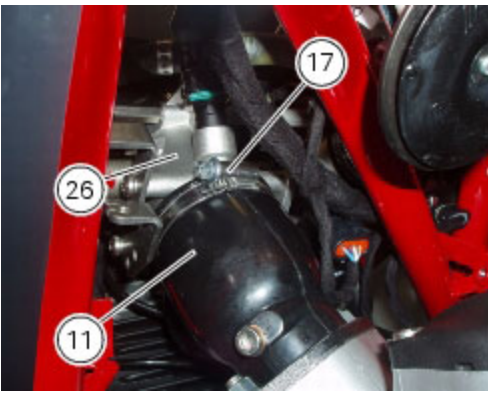


Locate the throttle body assembly (26) on the motorcycle.

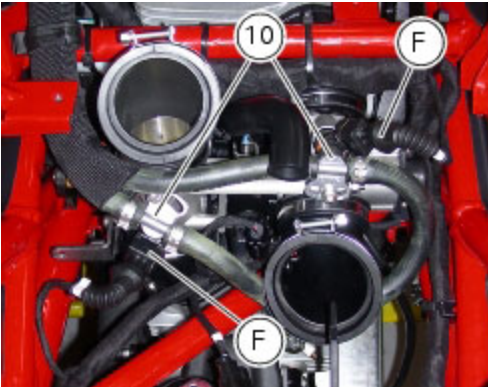
Fit the clamps (17) on the intake manifolds (11) and orient them as shown in the photos.

Orient the throttle body (26) with the race facing the right-hand side (relative to the direction of travel) and insert it fully into the intake manifolds.

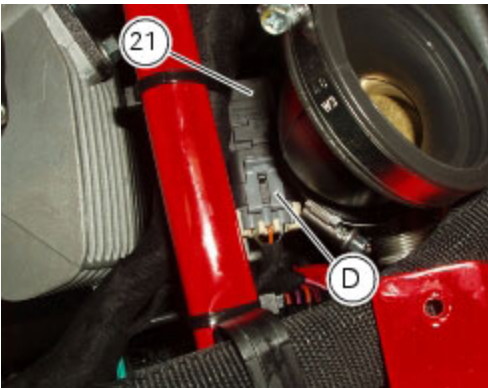
Tighten the clamps (17) to the specified torque (Sect. C 3, [Frame torque settings](#)), making sure that they do not protrude beyond the edges of the manifolds.



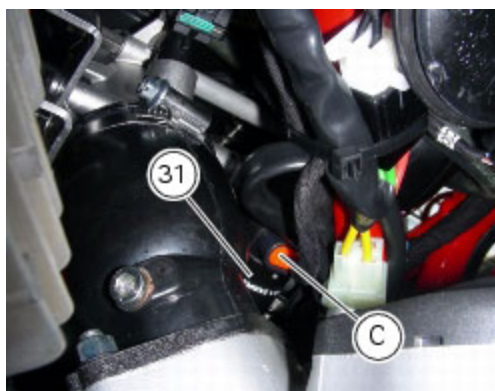
Reconnect the wiring connectors (F) to connect the main wiring loom to the injectors (10).



Install the air pressure sensor (21) to the throttle body, fit pressure sensor hoses to manifold and connect connector (D) to main wiring harness.



Connect the connector (C) connecting the main wiring loom to the stepper motor (31).



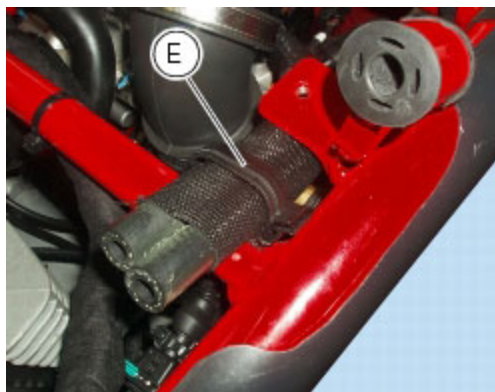
Connect the wiring connector (B) connecting the main wiring harness to the air temperature sensor (7).



Connect the connector (A) connecting the main wiring harness to the throttle position sensor (2).



Fasten the delivery and return hoses to the frame by means of the clip clamp (E).



Operation	Section reference
Connect the throttle cable	F 1, Refitting the throttle twistgrip
Refit the airbox	L 7, Refitting the airbox
Refit the battery	P 2, Refitting the battery

Connect the starter motor/solenoid cable	P 3, Refitting the starter motor
Connect the ECU connector	M 3, Refitting the electronic control unit
Refit the ABS control unit	G 9.2, Refitting the ABS control unit
Refit the fuel tank	L 2, Refitting the fuel tank
Refit the tank covers	E 2, Refitting the fuel tank fairings
Refit the seat	E 3, Refitting the seat

Positioning of the hoses on throttle body

